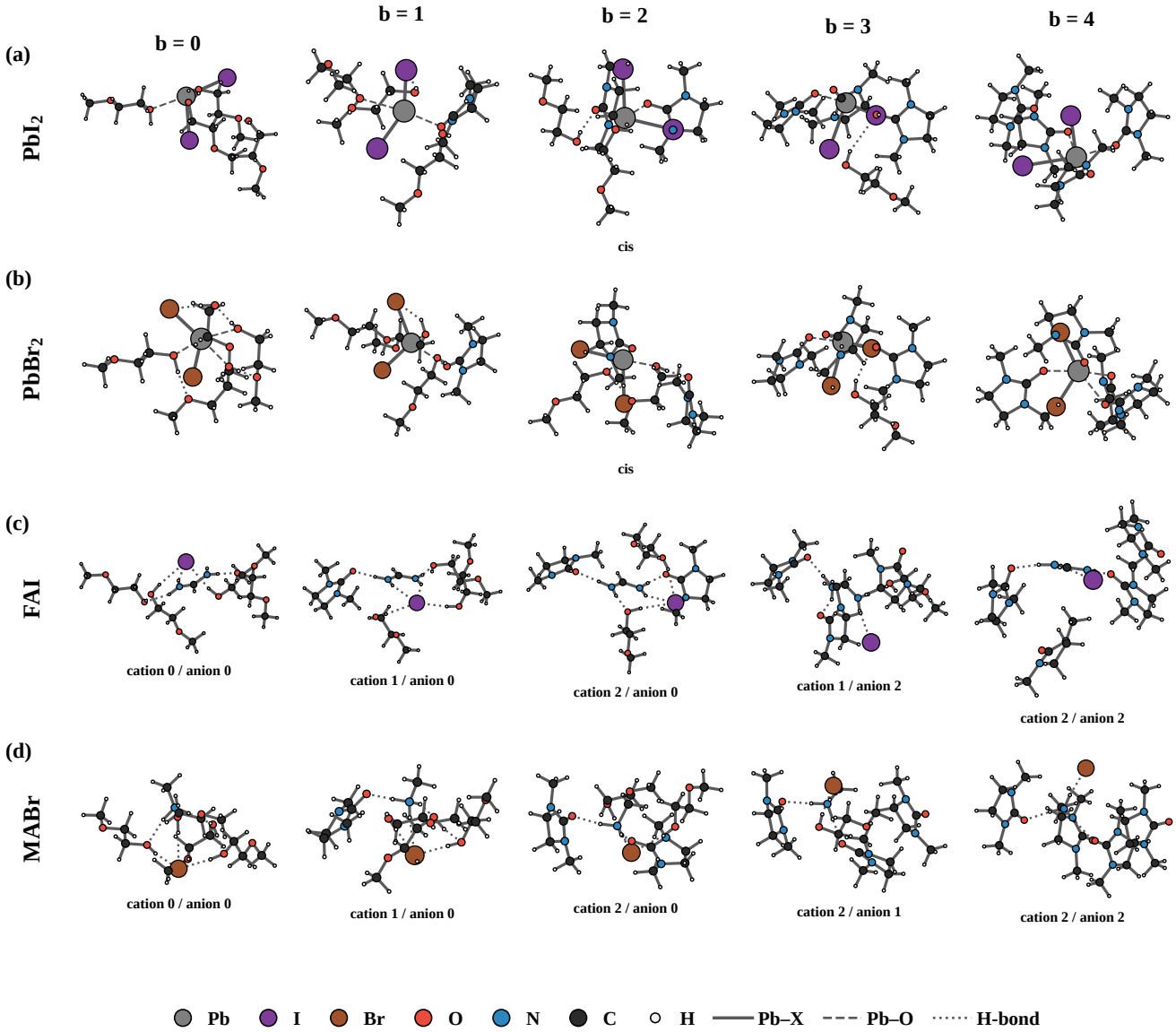

b = 0
b = 1
b = 2
b = 3
b = 4
(a)
PbI2
cis
(b)
PbBr2
cis
(c)
FAI
cation 0 / anion 0
cation 1 / anion 0
cation 2 / anion 0
cation 1 / anion 2
cation 2 / anion 2
(d)
MABr
cation 0 / anion 0
cation 1 / anion 0
cation 2 / anion 0
cation 2 / anion 1
cation 2 / anion 2
Pb
I
Br
O
N
C
H
Pb–X
Pb–O
H-bond